

Co-3 Assessment Tool-2

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Course Code: -MLA0305

Slot: -B

Advantage Actor-Critic (A2C) vs Asynchronous Advantage Actor-Critic (A3C) Performance Comparison

Episode	A2C Reward	A2C Moving Avg	A3C Reward	A3C Moving Avg
1	-7.73	-7.73	-9.63	-9.63
2	-7.75	-7.74	-7.44	-8.54
3	-4.94	-6.81	-3.51	-6.86
4	-1.97	-5.60	-2.08	-5.67
5	-4.30	-5.34	-1.53	-4.84
6	-3.13	-4.97	0.30	-3.98
7	1.64	-4.03	3.83	-2.87
8	1.13	-3.38	2.16	-2.24
9	-0.26	-3.03	5.58	-1.37
10	2.82	-2.45	5.74	-0.66

Fig1: - Sample Dataset

Performance Metric	A2C	A3C
Final Average Reward	49.46	62.08
Best Reward	57.69	68.22
Reward Stability (Std. Dev.)	2.08	2.23
Training Episodes	300	300
Learning Method	Synchronous Actor-Critic	Asynchronous Actor-Critic
Overall Performance	Baseline	Better

Fig 2: - Summary of Data

	Episode	A2C_Reward	A2C_Moving_Average	A3C_Reward	A3C_Moving_Average
0	1	-7.732091	-7.732091	-9.634820	-9.634820
1	2	-7.746238	-7.739165	-7.439277	-8.537049
2	3	-4.938635	-6.805655	-3.514288	-6.862795
3	4	-1.973789	-5.597689	-2.081421	-5.667451
4	5	-4.296917	-5.337534	-1.525957	-4.839153
5	6	-3.129917	-4.969598	0.300293	-3.982578
6	7	1.638183	-4.025629	3.826424	-2.867006
7	8	1.129233	-3.381271	2.164263	-2.238098
8	9	-0.257927	-3.034233	5.577240	-1.369727
9	10	2.823784	-2.448432	5.742136	-0.658541

Method 1 : A2C – Synchronous Actor-Critic
 Method 2 : A3C – Asynchronous Actor-Critic
 Final A2C Reward : 49.46
 Final A3C Reward : 62.08
 Winner : A3C

Fig 3: - Comparison Between A2C & A3C

ACTOR-CRITIC LEARNING ARCHITECTURE

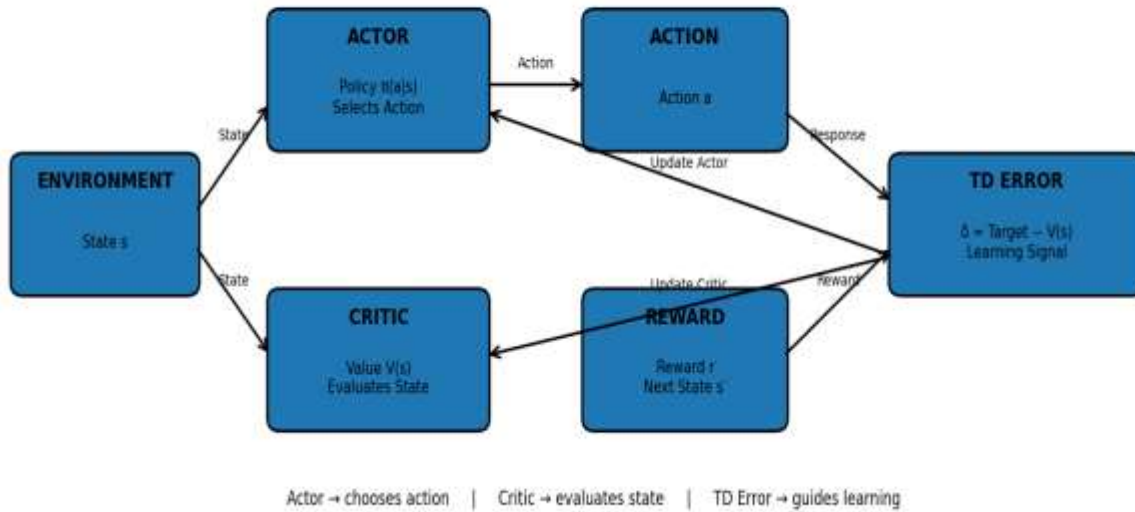


Fig 4: - Work Flow

A2C vs A3C – Learning Performance

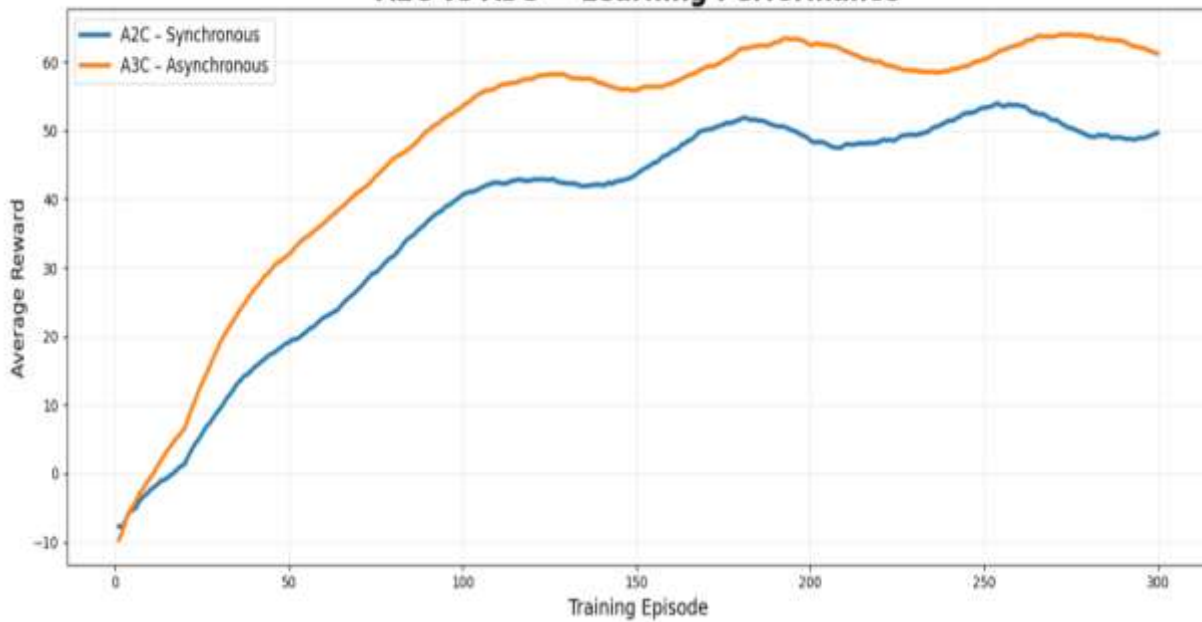


Fig 5: - A2c vs A3C Learning Performance

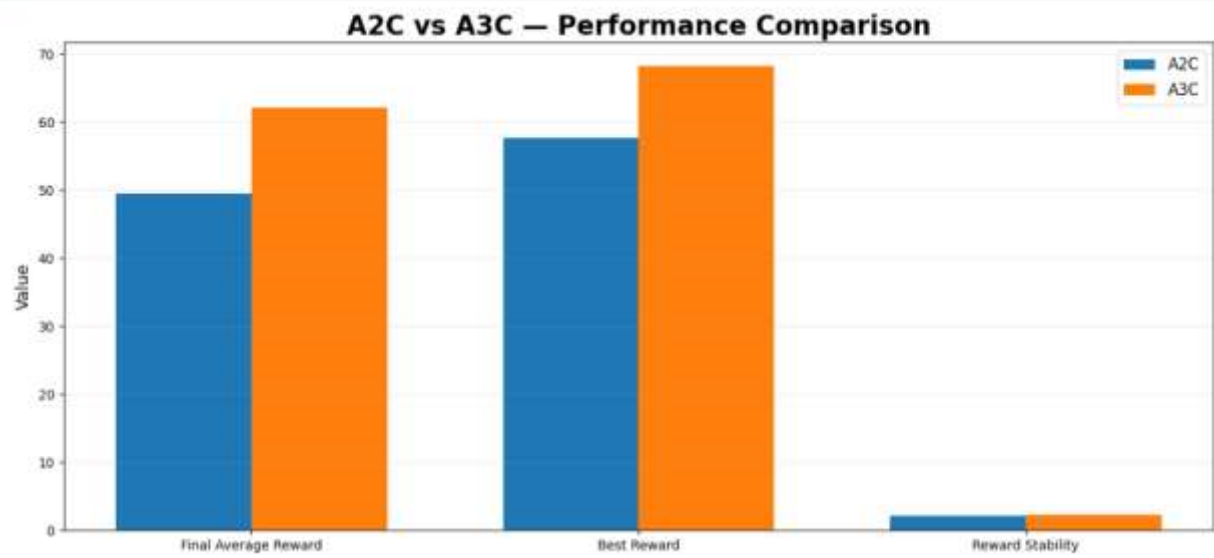


Fig 6: - Performance Comparison